

ISM4641: Python for Business Analytics

Summary of topics - Week1 through 2

Dr. Tim Smith

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Week 1: Introduction to Python and Business Analytics

- **Why Python for Business Analytics?**
 - Data as a strategic asset
 - Competitive advantage through data analysis
 - Industry trends (automation, AI)
- **Introduction to Google Colab**
 - What is Google Colab?
 - Notebook cells (code and text)
 - Basic Markdown formatting
 - Running code
 - Common shortcuts
- **Variables**
 - Naming rules and conventions
 - Assignment operator (=)
- **Data Types**
 - Integer (`int`)
 - Float (`float`)
 - String (`str`)
 - Boolean (`bool`)
 - Checking data types with `type()`
- **Operators**
 - Arithmetic operators (+, -, *, /, //, %, **)
 - Comparison operators (==, !=, >, <, >=, <=)
 - Logical operators (and, or, not)

- **Type Conversion**

- `int()`
- `float()`
- `str()`

- **Comments**

- Single-line comments (`#`)
- Multi-line comments (`"""Docstrings"""`)

- **Print function**

- Printing variables
- Printing text
- Printing multiple items
- Using `sep` and `end` parameters (implicitly)

- **Markdown**

- Headings
- Emphasis (italic, bold)
- Lists (unordered, ordered)
- Links
- Code (inline, block)

- **Input function** (implicitly, used in problems/assignments)

Week 2: Control Flow and Data Structures (Lists and Dictionaries)

- **Conditional Statements**

- `if`
- `elif`
- `else`

- **Loops**

- `for` loops
- `while` loops
- `break` statement (used in problems/assignments)
- Iterating through lists and dictionaries

- **Lists**

- Creating lists
- Accessing elements by index
- Modifying lists (adding, removing elements)
- Slicing lists

- List methods (e.g., `append()`, used in problems/assignments)
- **Dictionaries**
 - Creating dictionaries
 - Accessing values by key
 - Modifying dictionaries (adding, updating, removing key-value pairs)
 - Dictionary methods (e.g., `items()`, used in examples)
- **Input function**
- **f-strings** (implicitly, used in output formatting)
- **String methods** (e.g., `lower()`, used in problems/assignments)